A Later Negative Answer to Wang–Zhang Problem 4

Smooth-phase Stein–Wainger principal values

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Status. `literature_already_answered` (exact negative answer). This is a literature-identification packet, not a claim of a new solution.

Original question

Wang and Zhang ask in Section 4, Problem 4 (bottom of PDF page 14) of [1]:

> Let $\lambda \in \mathbb{R}$ and $\psi \in C^\infty(\mathbb{R})$. Is
>
> $$m_\psi(\lambda) = \mathrm{p.\,v.} \int_{-1}^{1} e^{i\lambda\psi(t)} \, \frac{dt}{t}$$
>
> bounded on $\mathbb{R}$?

Decisive later answer

Zhang and Zhu explicitly state that they resolve Wang–Zhang's problem [2, abstract and §1.2]. Their Theorem 1.2 says that every real $\psi \in C^\infty([-1, 1])$ satisfies

$$m_\psi(\lambda) = o(\log \lambda),$$

but this is sharp in the following strong sense: for every integer $k \geq 2$ there exist a real smooth phase ψ and a sequence $\lambda_n \to \infty$ such that

$$|m_\psi(\lambda_n)| \asymp \frac{\log \lambda_n}{\log^{(k)} \lambda_n}.$$

In particular, m_ψ is unbounded, so the answer to Problem 4 is *no*. A smooth phase on the compact interval extends smoothly to $\mathbb{R}$ without changing this integral. The supporting paper also says verbatim in its introduction that its results give a negative answer to "Wang–Zhang [28, Problem 4 in Section 4]." Thus the identification is explicitly recognized by the supporting authors.

Scope and search evidence

The later theorem settles exactly the scalar boundedness question selected from the source paper; the source paper's other geometric restriction problems are not addressed by this packet. A bounded search on 19 July 2026 used arXiv ids 1606.09346 and 2603.23238, the exact displayed principal-value formula, the source title and authors, and "Stein–Wainger oscillatory integral." It found the decisive 2026 arXiv preprint and an unanswered 2018 MathOverflow instance of the same bounded-interval question, but no earlier separate paper explicitly resolving the Wang–Zhang formulation.

Human-review recommendation

Confirm the source location and compare Theorem 1.2 and the explicit identification in Section 1.2 of arXiv:2603.23238. No mathematical inference beyond smooth extension from $[-1, 1]$ to $\mathbb{R}$ is needed.

References

[1] X. Wang and C. Zhang, *Sharp endpoint estimates for eigenfunctions restricted to submanifolds of codimension 2*, Adv. Math. 386 (2021), 107835; arXiv:1606.09346.

[2] C. Zhang and Z. Zhu, *Maximal growth of the Stein–Wainger oscillatory integral*, arXiv:2603.23238v1 (2026).